

WASP-10b

R-0733

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_221107 · created 2026-07-28T01:14:19+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459890.7485 +/- 0.0051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.161 +/- 0.048
Transit depth (Rp/Rs)^2	2.58 +/- 1.53 [%]
Orbital Inclination (inc)	87.54 +/- 1.54
Airmass coefficient 1 (a1)	0.998 +/- 0.038
Airmass coefficient 2 (a2)	0.008 +/- 0.026
Scatter in the residuals of the lightcurve fit is	3.82 %
Best Comparison Star	#2 - [361, 348]
Optimal Aperture	3.09
Optimal Annulus	14.09
Transit Duration (day)	0.082 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.161 ± 0.048 vs 0.1592 (deviation 0.04σ)
Duration fit vs published	0.082 d vs 0.0946 d (deviation 0.74σ)
Mid-time O–C	0.2 min
Depth SNR	3.4
Residual scatter	3.82 %

Light curve

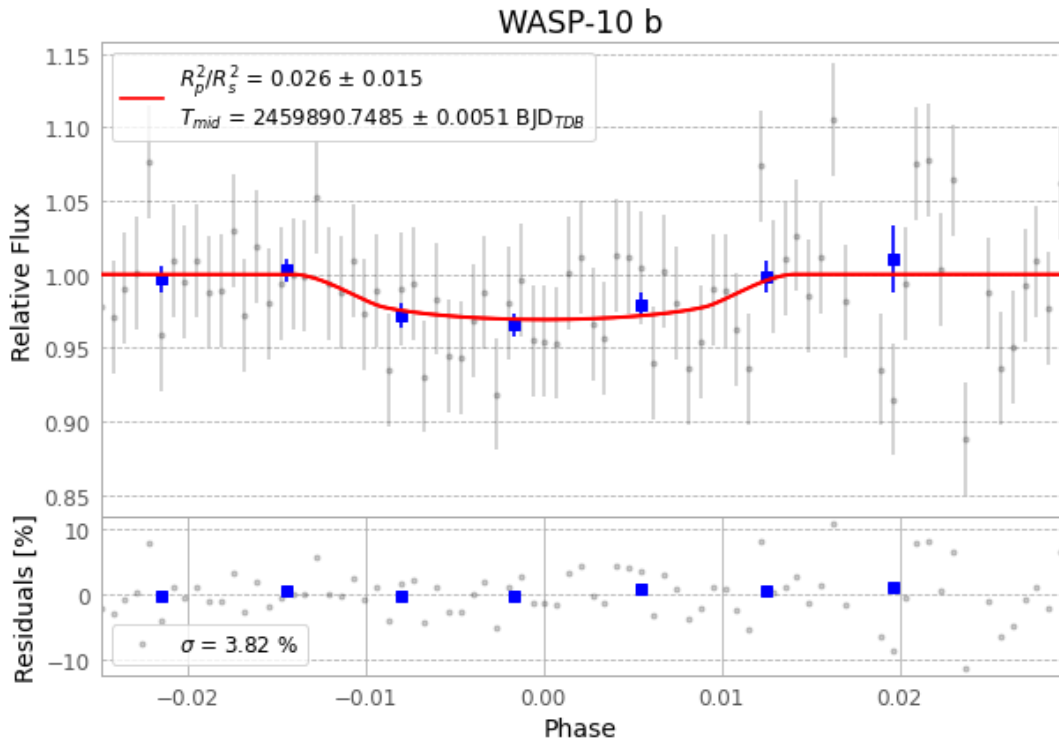


Figure 1 — FinalLightCurve_WASP-10 b_2022-11-06.png (SHA-256 32e38fa765dfb972...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [366, 222], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4215d39c5b90844e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

81 FITS frames (53 MB), WASP-10221107040015.FITS → WASP-10221107080015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-221107.log (e2ff1d3fd198...), FinalLightCurve_WASP-10 b_2022-11-06.png (32e38fa765df...), FinalLightCurve_WASP-10 b_2022-11-06.csv (57297da5f7d6...)

Compute resources

Wall time 375.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 01:14Z.